Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

SI TAC Production Centre

Arctic Ocean - Sea and Sea-Ice Surface Temperature Products

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008

SEAICE_ARC_PHY_SST_IST_L3S_NRT_011_022

Issue: 3.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	December 2014	All	First draft version, based on the template from the OSI-TAC Wiki	J. Høyer	Bruce Hackett
1.1	March 2015	All	Revised version based upon feedback from QuARG	J. Høyer	Bruce Hackett
1.2	September 2015	All	Change format to fit CMEMS graphical rules. Introduce references to CMEMS.	B. Hackett	L. Crosnier
1.3	January 2017	All	Updated with SST results using the version 3.0 SST bias correction method	J. Høyer	
1.4	April 2019	All	Updated with the inclusion of the Sentinel 3 SLSTR SST data	J. Høyer	Mercator Ocean
1.5	January 2020	All	Updated with the inclusion of Sentinel 3B, and VIIRS_NPP SST/IST and NOAA 20 SST data	J. Høyer	Mercator Ocean
1.6	September 2021	All	Introduced new consistent sea ice product.	I. Karagali, J. Høyer, P. Englyst	
2.0	June 2023	All	New template, introduce validation of IST with SIMB3, introduce validation of error analysis field	I. Karagali, M. Barfod Suhr, P. Englyst	C. Wettre
2.1	July 2024	All	Sea Ice Concentration input data	I. Karagali, P. Englyst, I. Lundtorp Olsen	C. Wettre
3.0	May 2025	All	New template, validation of L3S product	I. Karagali, W. Kolbe, G. Gacitúa	C. Wettre

ACRONYM TABLE

Acronym	Description
Argo	International profiling float network
AVHRR	Advanced Very High Resolution Radiometer
CCI	Climate Change Initiative
CDR	Climate Data Record
C3S	Copernicus Climate Change Service
DMI	Danish Meteorological Institute
EAN	Estimated Accuracy Number
EiS	Entry into Service
ESA	European Space Agency
EUMETSAT	European Meteorological Satellites
GCOS	Global Climate Observing System
GHRSST	Group for High Resolution Sea Surface Temperature
ICDR	Interim Climate Data Record
IST	Sea-Ice Surface Temperature
MetOp	Meteorological Operational Satellites
NASA	National Aeronautics and Space Administration
NOAA	National Oceanic and Atmospheric Administration
NPP	National Polar-orbiting Partnership
NRT	Near Real Time
OI	Optimal Interpolation
OSTIA	Operational Sea Surface Temperature and Ice Analysis
PMW	Passive Microwave
SAF	Satellite Application Facility
SI	Sea Ice
SIC	Sea Ice Concentration
SLSTR	Sea and Land Surface Temperature Radiometer
SST	Sea Surface Temperature
TAC	Copernicus Marine Service Thematic Assembly Centre
TIR	Thermal Infrared
VIIRS	Visible Infrared Imaging Radiometer Suite

KEY DEFINITIONS

Sea Surface Temperature	The temperature of the ocean water at 20 cm depth.
Sea-Ice Surface Temperature	The temperature of the sea-ice surface.
Thermal skin layer	The layer within which the long-wave radiation penetrates the ocean, typically less than 1mm, where ocean-atmosphere heat exchange takes place and temperature is measured using Infra-red sensors.
$SST_{sub\text{-}skin}$	Sea Surface Temperature at the base of the skin layer.
SST_{skin}	Sea Surface skin temperature, which is the mean temperature of the thermal skin layer.
$SST_{foundation}$	Sea Surface Temperature at the base of the thermal layer, the diurnal thermocline, which is considered representative of night-time and/or well mixed conditions, i.e free of any diurnal signal.
Sea Ice Concentration	Fraction (percentage) of the grid cell where the sea surface temperature is quantified containing sea-ice.
Sea Ice	Ocean surface covered by ice (>70%). For these grid cells the surface temperature is classified as sea-ice surface temperature (IST).
Level-2 Pre processed (L2P) data product	Geophysical variables derived from Level 1 source data on the Level 1 grid (typically the satellite swath projection). Ancillary data and metadata added following GHRSST Data Specification.
Level-3 Uncollated (L3U) data product	L2 data granules remapped to a regular latitude/longitude grid without combining observations from multiple source files. L3U files will typically be “sparse”, corresponding to a single satellite orbit.
Level-3 Collated (L3C) data product	SST observations from a single instrument combined into a space-time grid. In this project, a typical L3C file may contain all the observations from a single instrument in a 24-hour period.
Level-3 Super-Collated (L3S) data product	SST observations from many instruments combined into a space-time grid. In this project, a typical L3S file may contain all the observations from all available instruments in a 24-hour period.
Level 4 (L4) data product	SST observations from multiple instruments using an analysis system (e.g. optimal interpolation) to produce a gridded, gap-free product.

RELEVANT DOCUMENT LIST

CMEMS-SEAICE-SQO-011-008-022	The Synthesis Quality Overview document is a short summary of the quality information presented here. It can be found in the dedicated product page[1] under Documentation[2]
CMEMS-SEAICE-PUM-011-008-022	The Product User Manual presents the product details along with file structure. It can be found in the dedicated product page under Documentation[3]

[1] https://data.marine.copernicus.eu/product/SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008/description
[2] https://documentation.marine.copernicus.eu/SQO/CMEMS-SEAICE-SQO-011-008.pdf
[3] https://documentation.marine.copernicus.eu/PUM/CMEMS-SI-PUM-011-008.pdf

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008: The L4 Sea and Sea-Ice Surface Temperature product for the Arctic Ocean (gap-free), at 0.05° resolution, including surface temperature of the open ocean (SST), sea ice (IST) and marginal ice zone (MIZ).

SEAICE_ARC_PHY_SST_IST_L3S_NRT_011_022: The L3S Sea and Sea-Ice Surface Temperature product for the Arctic Ocean (with gaps due to missing data), at 0.05° resolution, including surface temperature of the open ocean (SST), sea ice (IST) and marginal ice zone (MIZ).

I.2 Summary of the results

The quality of the L4 Sea Surface Temperature (SST) and Sea-Ice Surface Temperature (IST) has been assessed for 2012 and 2019-2022 for the IST part and for the period October 2016 to January 2017 for the SST part. The validation against in situ observations for the SST has been performed using drifting buoys. Validation of the IST component was performed using Ice Mass Balance (IMB) buoys from the NAACOS[4] project for the period August-December 2012 and Sea Ice Mass Balance Buoys (SIMB3) buoys from 2019 to 2022.

The quality of the L3S Sea Surface Temperature (SST) and Sea-Ice Surface Temperature (IST) has been assessed for the year 2023 for both the IST and SST components. The validation against in situ observations for the SST has been performed using drifting buoys while validation of the IST component was performed using Ice the Sea Ice Mass Balance Buoys (SIMB3) buoys.

The results for both the L4 and L3S product are summarised below.

Sea Surface Temperature: For the L4 product a mean difference of -0.10°C and a standard deviation of 0.55°C when compared against drifting buoys is reported.

For the L3S product, a mean difference of 0.10°C and a standard deviation of 1.20°C when compared against drifting buoys in June-October 2024 is reported.

Sea-Ice Surface Temperature: For the L4 product, mean difference of -4.87°C and a standard deviation of 2.88°C when compared to IST measurements from the NAACOS IMB buoys and a mean difference of -2.81°C and standard deviation of 3.94°C using air temperature measurements from SIMB3 buoys.

For the L3S product, a mean difference of -4.79°C and a standard deviation of 2.94°C when compared to the SIMB3 buoys in June-September 2024.

Important information or **noticeable event**: During the course of the operational system analysis, new sensors have been introduced (SST from SLSTR on Sentinel-3B, SST and IST from VIIRS on Suomi-NPP and SST from VIIRS on NOAA20). Furthermore, changes in the sea ice concentration input data have been made throughout the course of the operational system analysis aiming to improve the characterization of open water, marginal ice zone and sea-ice covered regions (see sections IV2 and VI3).

[4] https://orbit.dtu.dk/en/projects/north-atlantic-arctic-coupling-in-a-changing-climate-impacts-on-o

I.3 Estimated Accuracy Numbers

A summary of the estimated accuracy numbers, derived from the validation of the L4 product and the L3S product against drifting buoys for the SST component and SIMB3 buoys for the IST component is presented in Table 1. Note that for the L4 product also NAACOS IMB buoys were used to validate the IST component.

Table 1. Table of estimated accuracy numbers (where variables, units and numbers are given).

Variable	Units	In Situ	Metrics	Values of metric
Sea Surface Temperature (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_022)	°C	Drifting buoys	mean bias, standard deviation, relative standard deviation, Number of observations (Nobs)	-0.10, 0.55, 0.56, 22544
Sea Surface Temperature (SEAICE_ARC_PHY_SST_IST_L3S_NRT_011_008)	°C	Drifting buoys	mean bias, standard deviation, relative standard deviation, Number of observations (Nobs)	0.10, 1.20, 0.55, 30595
Ice Surface Temperature (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008)	°C	NAACOSIMB buoys	mean bias, standard deviation, relative standard deviation, Number of observations (Nobs)	-4.87, 2.88, 5.66, 2110
Ice Surface Temperature (SEAICE_ARC_PHY_SST_IST_L3S_NRT_011_022)	°C	SIMB3 buoys	mean bias, standard deviation, relative standard deviation, Number of observations (Nobs)	-4.79, 2.94, 2.64, 156

II PRODUCTION SYSTEM DESCRIPTION

Production centre name: Danish Meteorological Institute, DMI, Denmark

Production system name: Arctic Sea Surface and Sea Ice Surface Temperature L3S and L4 Analysis (Copernicus Marine Environment Monitoring Service names: SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008, SEAICE_ARC_PHY_SST_IST_L3S_NRT_011_022)

Description: The Danish Meteorological Institute Optimal Interpolation (DMIOI) analysis system is run daily in the operational suite at DMI. Figure 1 shows the different steps for the creation of the daily L3S and L4 SST fields covering the Arctic Ocean. The final level 4 analysis product is a merged and interpolated daily field with a 0.05° resolution in latitude and longitude, and covers surface temperatures in the ocean, the sea ice, and the marginal ice zone. The optimal interpolation (OI) method used to construct the merged and gap free SST/IST analysis is taken from the high latitude SST DMI processing scheme described in Høyer and She (2007) and Høyer et al. (2014). The OI method works with anomalies from a first guess field.

In the current approach, a persistence-based method is applied, which uses the previous analysis field as the first guess field. The SST and IST observations from the last 36 hours are therefore interpreted as anomalies with respect to the first guess field. The OI method operates at each grid point and finds the solution that has the lowest errors, given statistical input, such as a first guess error variance, error covariance functions and uncertainties on the individual observations. Due to the different physical conditions for ocean and sea ice surface temperature variability, separate statistics have been derived for the open ocean and the sea ice covered regions. The SST first guess variance and error covariance were very similar to the ones that were presented in Høyer and She (2007) and Høyer et al. (2014). Over sea ice, the first guess IST variability was derived using one year of MetOp Advanced Very High Resolution Radiometer (AVHRR) L3 aggregated observations. The previous day L4 field was subtracted from the L3 IST observations and a spatial 2-dimensional field of standard deviations were calculated for one year of anomalies. The first guess field and the error covariance used in the MIZ (Marginal Ice Zone) is a weighted linear combination of the open waters and the ice values, where the ice concentration was used as the weighting factor. In this context, the MIZ is defined as areas where the ice concentration is between 15% and 70%. The search radius for the OI method is set to 100 km and the maximum number of satellite observations included in the optimal estimation is 16. The average number of satellite observations included in the analysis is from 15.9 to 16, indicating good satellite data coverage within the search limits.

The SST bias correction method corrects the available MetOp AVHRR and Sentinel 3A+3B Sea Land Surface Temperature Radiometer (SLSTR) SST observations against those from the Visible/Infrared Imaging Radiometer Suite (VIIRS) on Suomi National Polar-orbiting Partnership (NPP) and National Oceanographic and Atmospheric Administration (NOAA) 20 platform. The temporal window for the comparison is seven days and the bias fields are smoothed over 500 km. See Høyer et al., 2014 and the Product User Manual for more information.

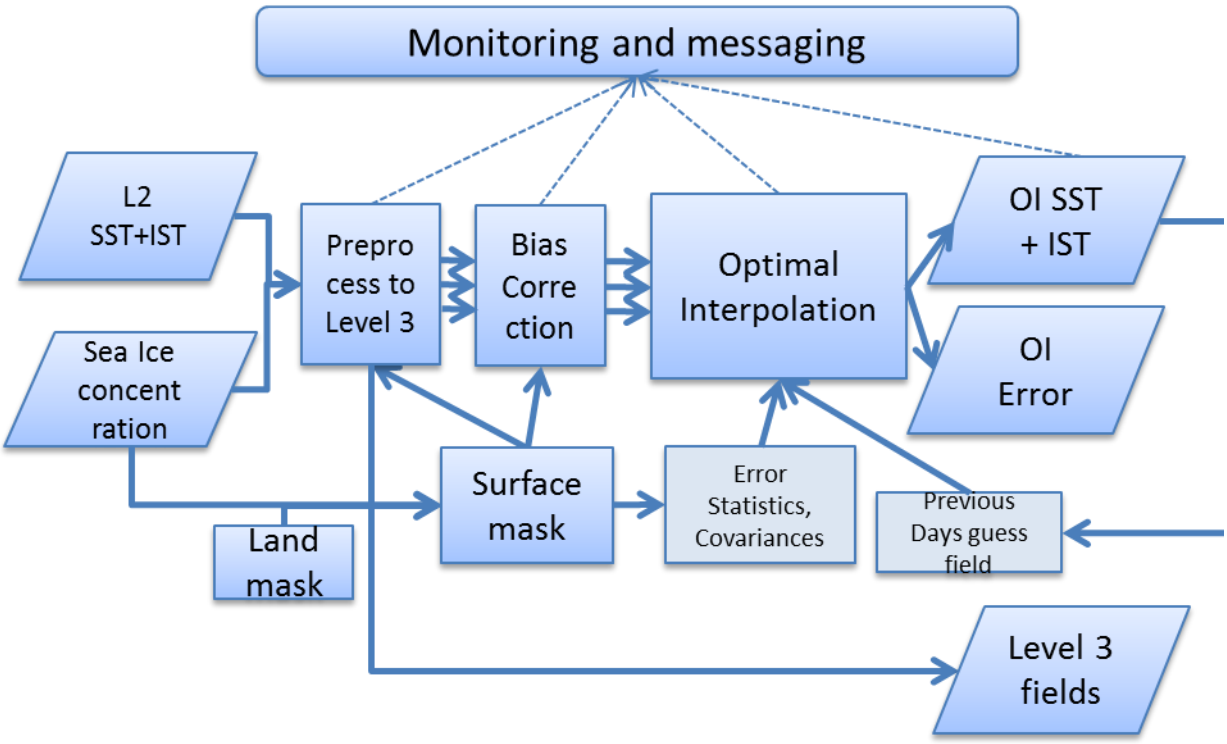

Figure 1. Schematic diagram of the operational processing chain of the Optimal Interpolation (OI) system at DMI.

III VALIDATION FRAMEWORK

The assessment of the performance of the subsystem is carried out as specified following a scientific qualification plan. This implies that the satellite SST observations are validated against drifting buoy observations. The availability of in situ ice surface temperature observations to be used for validation is very limited and the calibration period August-December 2012 was chosen due to the availability of ice surface temperature observations from NAACOS ice mass balance buoys that were deployed in August 2012. In addition, the near-surface air temperatures from Sea Ice Mass Balance Buoys (SIMB3) are used for validation of the IST part for the years 2019-2022.

For the L3S product, released later than the L4 product, drifting buoy observations and SIMB3 observations from June to October 2024 were used for validation of the SST and IST components.

III.1 Validation process when comparing to drifters

The observations form ICOADS[5] drifting buoys are used for the L4 SST validation. The positions of the SST matches against ICOADS observations for October 2016 - January 2017 are shown in Figure 2.

For the L3S validation, taking place for June-October2024, drifting buoy observations from the SST CCI Independent Reference Data Set (SIRDS) have been used.

[5] https://icoads.noaa.gov/

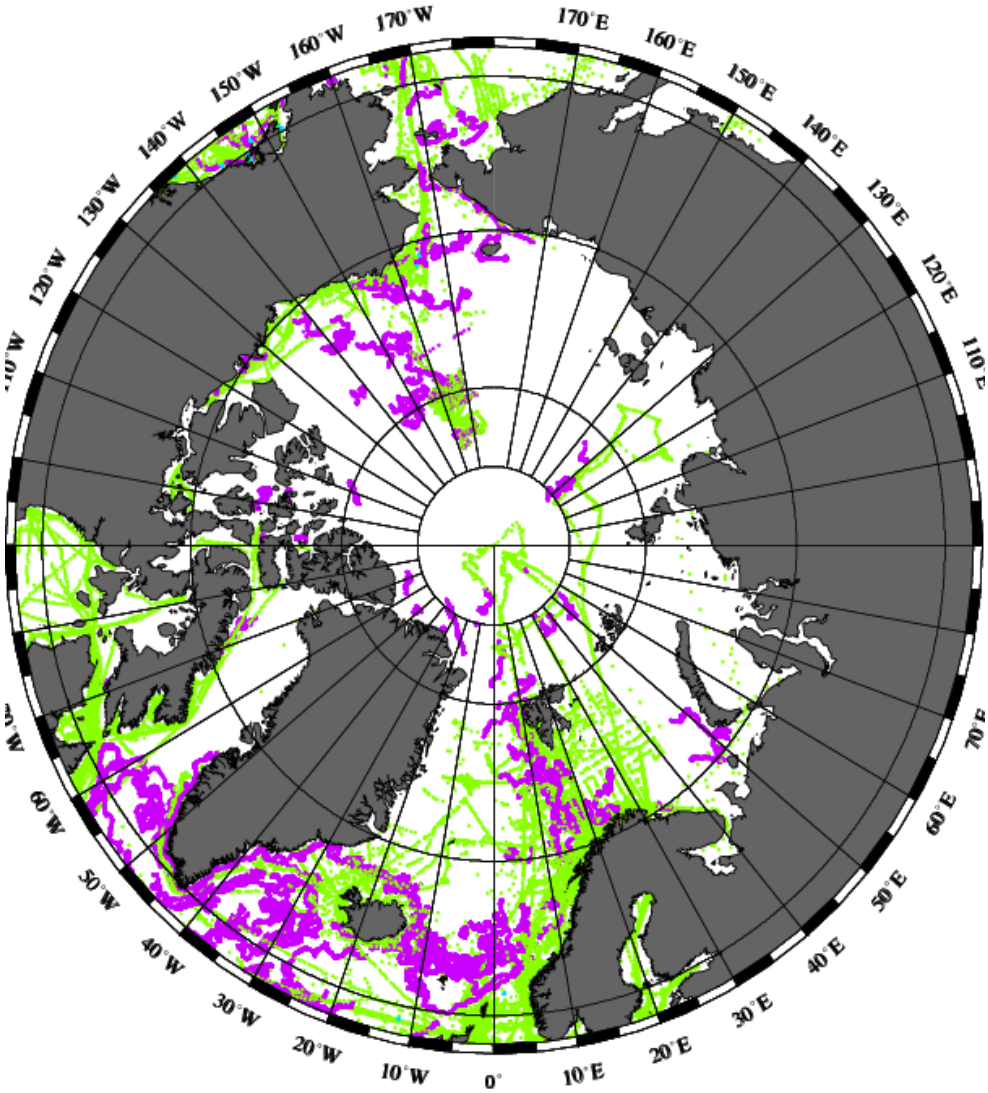

Figure 2. Positions of the L4 SST and ICOADS matches. The colours represent the type of observations, where drifting buoys are pink. Green lines indicate ship observations, but they were not used for validation.

III.2 Validation process when comparing to Ice Mass Balance Buoys

As part of the Danish research project North Atlantic - Arctic coupling in a changing climate: impacts on ocean circulation, carbon cycling and sea-ice (NAACOS), eight ice mass balance buoys were deployed from the Icebreaker ODEN in August 2012. The buoys were 3-4 m long thermistor strings with temperature sensors every 2 cm. Holes were drilled all the way through the Multiyear Ice floes and the IMB recorded the temperature of the multiyear ice floe, until it broke up or the buoy failed. Out of the eight buoys deployed, four were functioning throughout the calibration period. The Air/snow or Air/Ice interfaces were determined by manual inspection of the temperature records on the sensors. The interface temperature has been used in this study as the ice surface temperature. The ice surface temperature thus includes some uncertainty as to where the observation is from in relation to the surface. The observations cover the period August-December 2012, and the spatial coverage of the IMB buoys is shown in Figure 3.

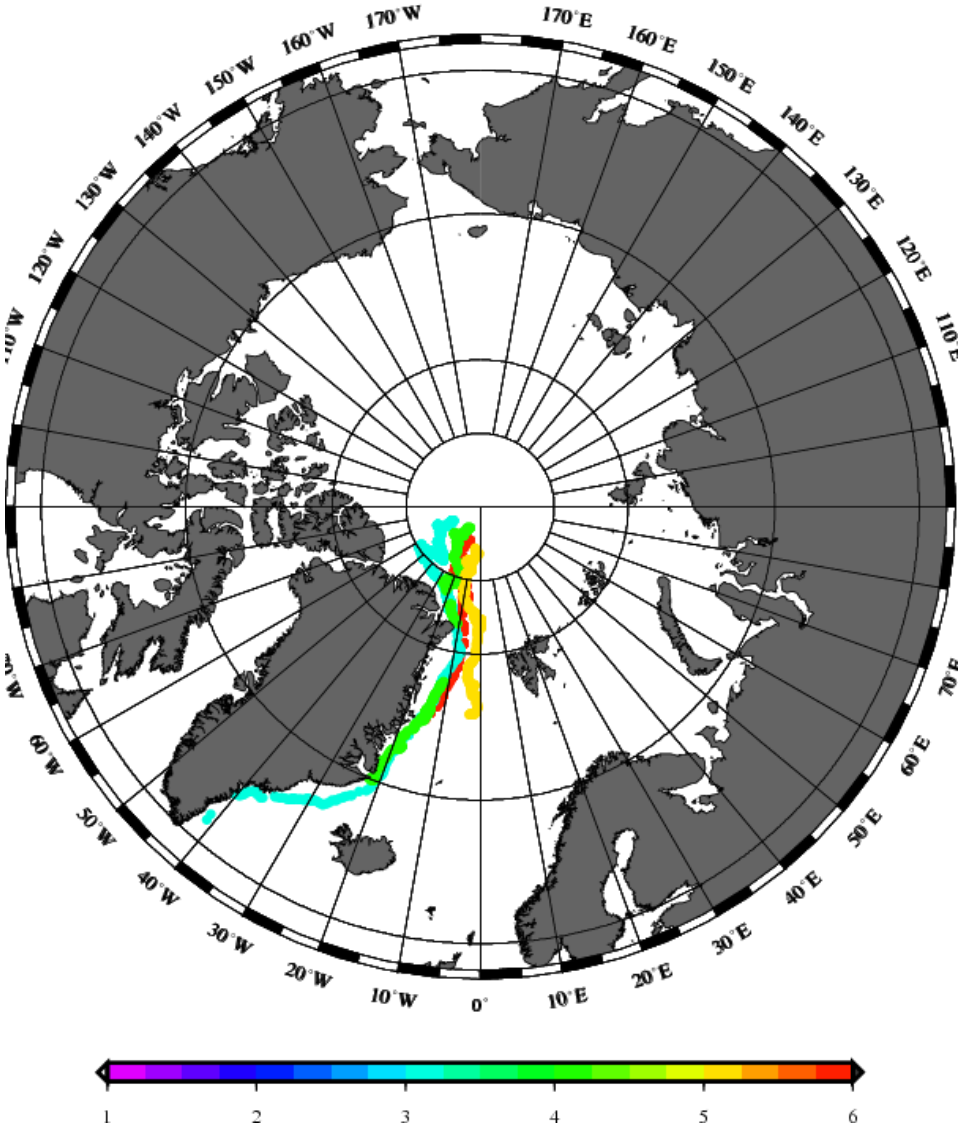

Figure 3. Positions of the NAACOS ice mass balance buoys used for the IST validation. The colour represents the buoy number (3-6). Buoys 1 and 2 are not shown because they were not relevant for the validation of the L4 IST component.

Another type of Ice Mass Balance buoy used for a limited period was the Alfred Werner Institute (AWI) Ice mass balance buoys. These were based on autonomous sea ice measurements (ice surface temperatures) and were obtained from the Meereis Portal[6] (grant REKLIM-2013-04), see Nicolaus et al. (2017).

[6] http://www.meereisportal.de

III.3 Validation process when comparing to near-surface air temperatures from SIMB3 buoys

Near-surface air temperature (Tair, approximately 1.2 m height) observations from Sea Ice Mass Balance Buoys (SIMB3) are used to validate the IST. The data is found at the Cryosphere Innovation website[7]. The positions of all the matches against SIMB3 observations for the years 2019-2022, used for the L4 validation, are shown in Figure 4.

For the L3S validation, SIMB3 buoys for June-September 2024 were used and the positions are shown in Figure 5.

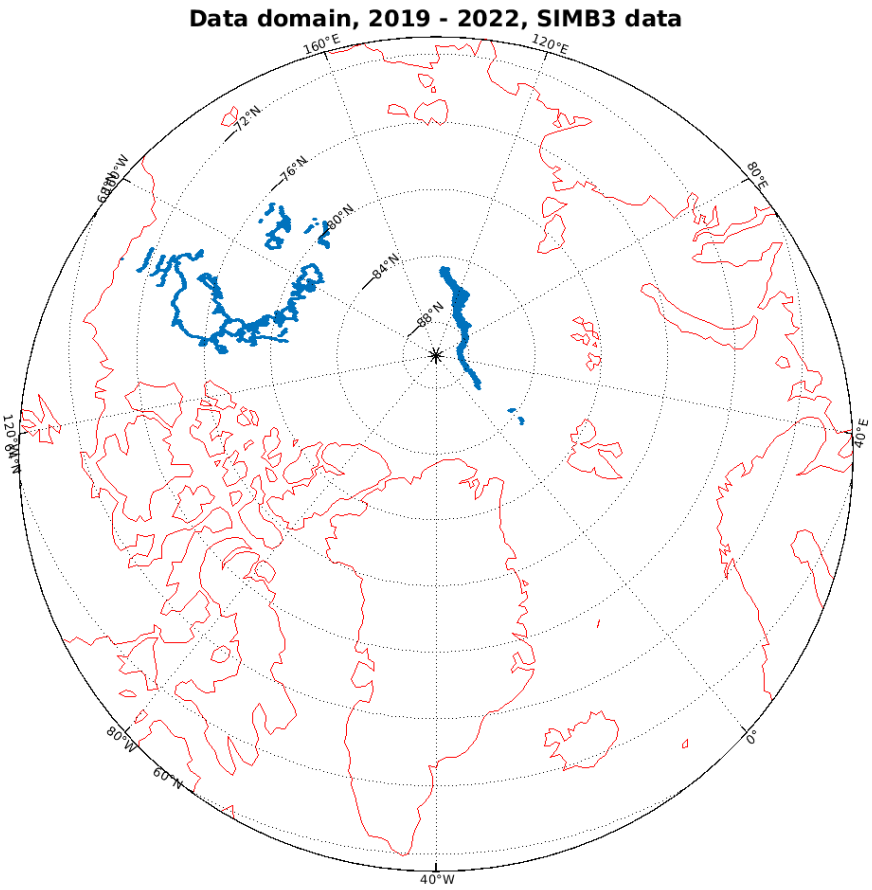

Figure 4. Positions of the L4 and SIMB3 matches for the years 2019-2022. The coloured lines indicate the tracks of the drifting buoys.

[7] https://www.cryosphereinnovation.com/data/

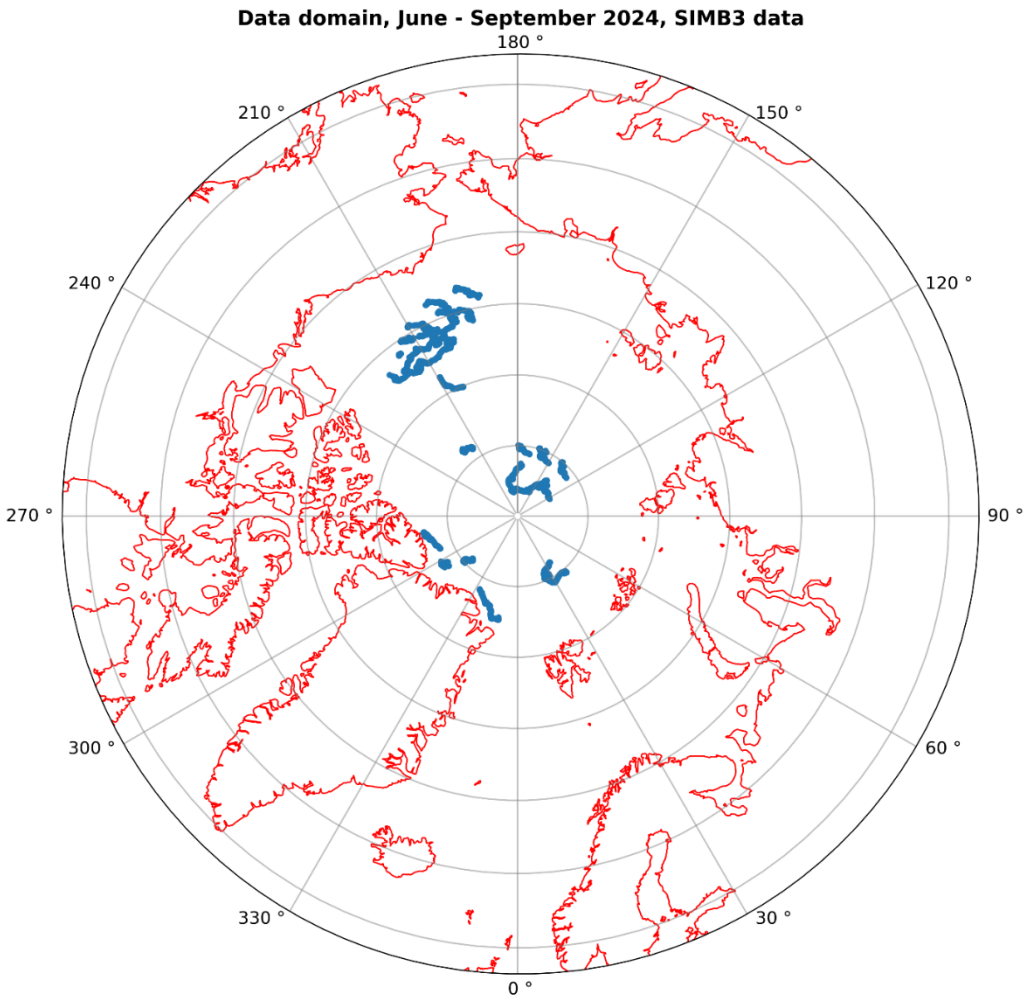

Figure 5. Positions of the SIMB3 observations for June-September 2024. The coloured lines indicate the tracks of the drifting buoys.

III.4 Validation process when lacking in-situ and or satellite data

No validation takes place.

IV VALIDATION RESULTS

IV.1 Validation of SST and IST

The validation of the L4 SST and IST analysis product is given in Table 2, where the in-situ observations are averaged in the same way as for the satellite products. Note that the IST validation results cover a period from August-December 2012, while the SST validation results cover the period from October 2016 to January 2017.

Furthermore, the validation of the L3S product for June-October2024 both for the SST component (using drifting buoys) and the IST component (using SIMB3 buoys) is also included. Note that due to the gaps of the L3S product, as it is meant to be the super-collated field of all the existing satellite observations that have passed the quality control, fewer match-ups with the in situ observations are to be expected. Note also that since the 2024 validation mainly covers summer, higher mean differences and standard deviations are expected compared to full year and winter validation periods.

Table 2. Overall validation statistics for the Level 4 SST/IST and the L3 SST/IST using in situ observations from drifting buoys for the SST component and from IMB buoys and SIMB3 buoys for the IST component.

Variable	In Situ	Mean Bias	Standard deviation	Relative Standard Deviation	Correlation	Nobs
Sea Surface Temperature (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008)	Drifting buoys	-0.10	0.55	0.56	0.99	22544
Sea Surface Temperature (SEAICE_ARC_PHY_SST_IST_L3S_NRT_011_022)	Drifting buoys	0.10	1.20	0.55	-	30595
Ice Surface Temperature (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008)	NAACOS IMB buoys	-4.87	2.88	5.66	0.94	2110
Ice Surface Temperature (SEAICE_ARC_PHY_SST_IST_L3S_NRT_011_022)	SIMB3 buoys	-4.79	2.94	2.64	-	156

As the differences reported in Table 2 are defined between the satellite and in situ observations, some issues that contribute to elevated differences are related to:

- Satellite SST product is a skin SST product, whereas the drifting buoy observations are from ~20 cm depth.
- The ICOADS SST observations include observations from regions covered with sea ice.
- The snow/ice-air interface on the IMB observations has been determined by visual inspection for each record. An error in the interface can introduce a significant difference due to the large vertical gradients in the snow and ice.
- The use of air measurements to validate IST will introduce a difference due to the vertical stratification in the near surface temperature.

IV.2 Validation results when Sentinel 3B SST, VIIRS Suomi-NPP SST/IST and NOAA20 SST were introduced

New upstream satellite data were introduced in the processing chain during the year 2020. To assess the impact of the new input data, the updated L4 product including the new Sentinel 3B, VIIRS_NPP SST/IST and NOAA 20 SST observations was compared to the original processing chain (without the new input data) during the test period from October to December 2020. The results are summarised in Table 3 and it is found that the SST bias of the product using the new upstream satellite inputs improved from -0.22°C to -0.10°C, while the standard deviation was reduced from 0.50°C to 0.47°C thus supporting the upstream data changes.

For assessment of the improvements in the IST performance, it was decided to use daily averaged observations of near surface air temperature (Tair) from the AWI snow buoy observations, as no reliable radiometric skin observations were available during this period. The physical difference between the skin temperature (Tskin) and the 2m air temperature (Tair) introduces a difference in the comparisons, which can be several degrees and not related to the performance of the product. Despite the large differences, Table 3 shows improvements in the IST performance when the new input data are introduced, with a reduction of the bias by 0.82°C and of the standard deviation by 0.23°C.

Table 3. Overall validation statistics for comparisons of the updated L4 analysis system using the new input data streams and the original L4 system (without the new data streams), against in situ observations for the period 10/2020 – 12/2020.

Variable	In Situ	Mean Bias	Standard deviation	Nobs
Sea Surface Temperature (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008) original	Drifting buoys	-0.22	0.50	24138
Sea Surface Temperature (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008) updated with S-3B, VIIRS Suomi-NPP and NOAA20		-0.10	0.47	24135
Sea-Ice Surface Temperature (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008) original	AWI snow buoy observations	-6.97	3.12	626
Sea-Ice Surface Temperature (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008) updated with S-3B, VIIRS Suomi-NPP and NOAA20		-6.15	2.89	6266

The validation statistics can change throughout the year, whereas the new validation period only covers a few months. The original validation results (Table 2) are also reported, as they cover more months. The new version, including the Sentinel 3B, VIIRS_NPP SST/IST and NOAA 20 SST observations appears to have significant improvement on the bias for Oct-December and small improvement in the standard deviations.

IV.3 Validation when the updated sea ice concentration fields were introduced

The Ocean and Sea Ice Satellite Application Facility (OSI-SAF) releases the sea ice concentration product OSI-401b1 which includes the variable ice_conc_unfiltered; this is used in the product to classify ice covered points. Additional filtering criteria were applied to minimize spurious sea ice using the sea surface temperature from the Operational Sea Surface Temperature and Ice Analysis (OSTIA; Donlon, 2012), product SST-GLO-SST-L4-NRT-OBSERVATIONS-010-001. The OSI-SAF OSI-401b product has challenges in coastal regions due to the coarse resolution of the Passive Microwave footprint and land contamination to the signal, and for the part of the product covering the Baltic Sea and adjacent waters the 1 km CMEMS SIC product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004 (2012-present) is used.

This improvement was implemented in EiS November 2021 (see v1.6 of this QUID document) and prior to implementation, the impact of adding the filtering criteria to the sea ice concentration product was assessed for a 3-month period; results are presented in Table 4 where **Reference** version is the one using the standard SIC product and **Updated** version is the one using the SIC product with updated filtering criteria. The statistics for the analysed surface temperature variable are calculated by estimating the difference in the L4 SST between the **Updated** and **Reference** version. Statistics for the sea ice

concentration variable are also included, considering sea ice concentrations above 15%. The values in parentheses show the mean bias, standard deviation, and number of matches for the sea ice fraction variable when no threshold is applied. Almost no difference between the two versions is identified.

When comparing the SST component of the **Updated** and **Reference** version against drifting buoy observations (bottom two rows of Table 4), lower mean and standard deviation values are observed for the Updated version. A small difference in the number of matches, approximately 1000 lower for the **Updated** version, is associated with the new filtering criteria that lead to more ice-covered points, where no drifter data are available.

*Table 4. Comparison (i.e., difference) between the **Updated** (with improved filtering criteria for the SIC product) minus the **Reference** version of the L4 SST component for the 3-month period July-September 2020, using all grid points in the domain. The two bottom rows show the validation statistics when the SST component of the two versions was compared against drifting buoy observations.*

Product	**Variable**	**Mean Bias**	**Standard deviation**	**Nobs**
SEAICE_ARC_SEAICE_L4_NRT _OBSERVATIONS_011_008 **Updated-Reference**	SST (°C)	-0.01	1.21	56988160
	Sea Ice Fraction	0.0014 (0.0048)	0.0206 (0.0344)	149877329 (254162880)
SEAICE_ARC_SEAICE_L4_NRT _OBSERVATIONS_011_008 **Updated-Drifting buoys**	SST (°C)	-0.02	0.99	113499
SEAICE_ARC_SEAICE_L4_NRT _OBSERVATIONS_011_008 **Reference-Drifting buoys**	SST (°C)	-0.11	1.04	114480

IV.4 Validation of the analysis error field

Assessing the provided error estimates for the SST component of the L4 product was performed with the use of drifting buoys for the test year 2021. The availability of in situ observations is shown in Figure 6 for drifting buoys.

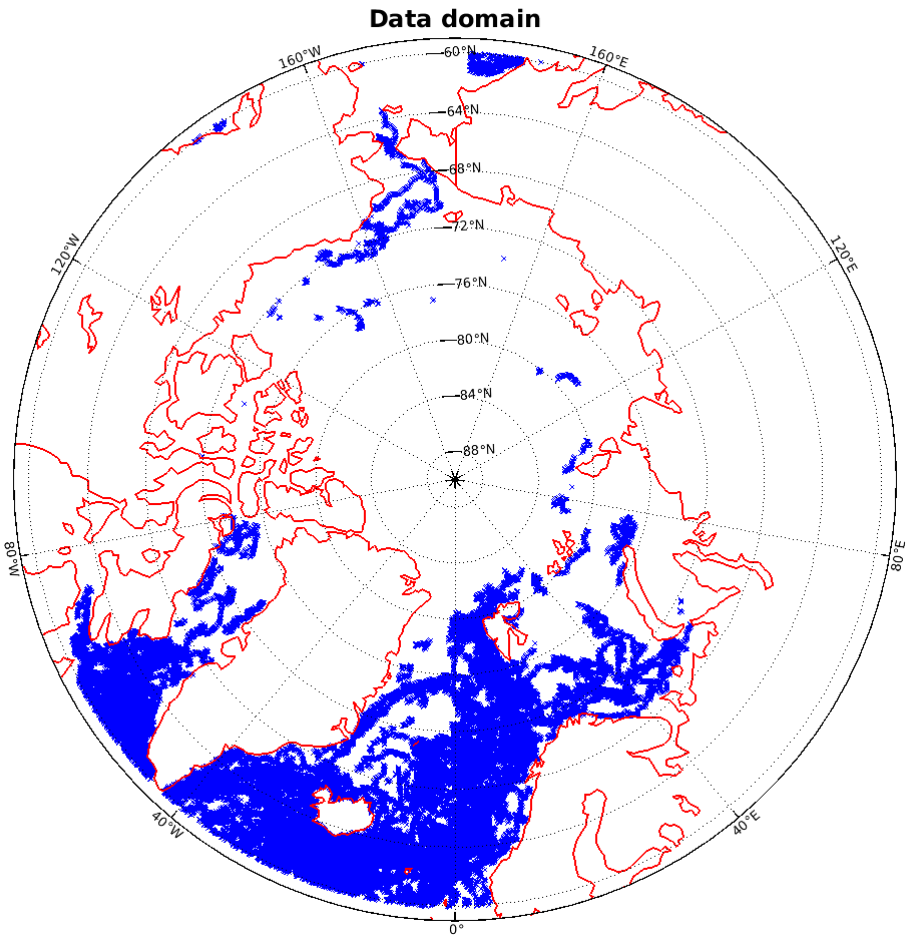

Figure 6. Map of drifting buoy tracks (blue points) available for the validation of the error estimates for 2021.

Error (or uncertainty) estimate validation was performed by assessing the estimated L4 uncertainty provided with the product versus the observed one, i.e. defined as the standard deviation of the L4 SST minus in situ. The validation of uncertainties using drifting in situ data is shown in Figure 7 (a) along with the number of match-ups used in (b). An overestimation of the modelled uncertainties that are provided with the daily L4 analysis is observed, manifested as a relatively stable standard deviation of L4 SST minus in situ (vertical black lines in (a)) although the modelled uncertainties increase from 0.2° and up to 1.6°. The theoretical curves which should be followed are indicated by the dashed lines, demonstrating that ideally as the modelled uncertainties increase (x-axis) the observed uncertainties (vertical lines) should also increase.

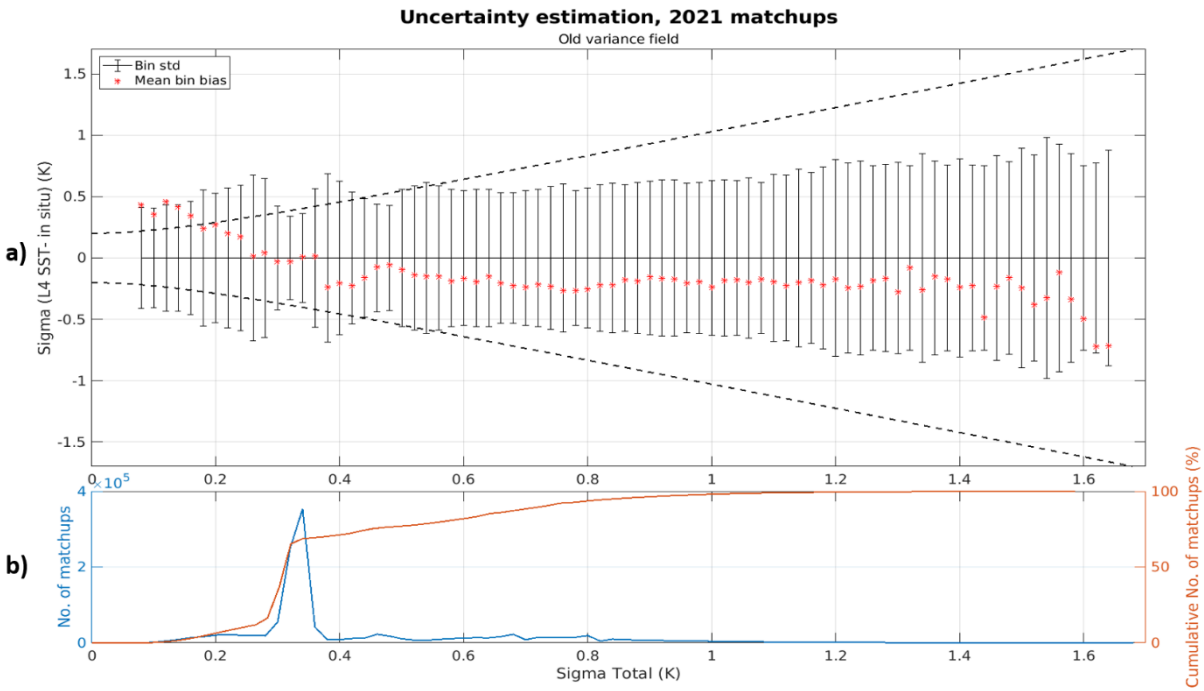

Figure 7. P (a): Uncertainties (x-axis) provided with the daily L4 analysis, against the estimated standard deviation of the L4 SST analysis minus in situ SST from buoys (y-axis), using drifters. The (b) shows the availability of match-ups (blue) and cumulative availability (orange).

The mean bias is shown as the red stars, indicating higher L4 SST minus in situ biases for the lower modelled uncertainties in the range between 0.1° and 0.25° indicating warmer L4 SSTs compared to the in situ. For modelled uncertainties of 0.4° and higher, biases turn negative and remain stable around -0.2° indicating warmer in situ SST compared to the L4 SST. Note that nonetheless, most data available for the uncertainty analysis (blue line in (b)) indicate zero biases for modelled uncertainties between 0.2° and 0.4°.

In order to reduce the modelled error estimates provided with the L4 daily analysis, an assessment of the 1st-guess variance used was performed. A new 1st-guess variance field was estimated using daily analyses fields from 2018 to 2021. The original (left) and new (middle) 1st-guess variance fields along with their difference (right) are shown in Figure 8. The new 1st-guess variance field was lower than the standard one used for the operational production, by up to 2° especially obvious in large parts of the study area including Greenland, the Kara, Laptev, Chukchi, and Beaufort Seas and along the Marginal Ice Zone. By assessing the differences, an overall reduction of the 1st guess variance was proposed for implementation.

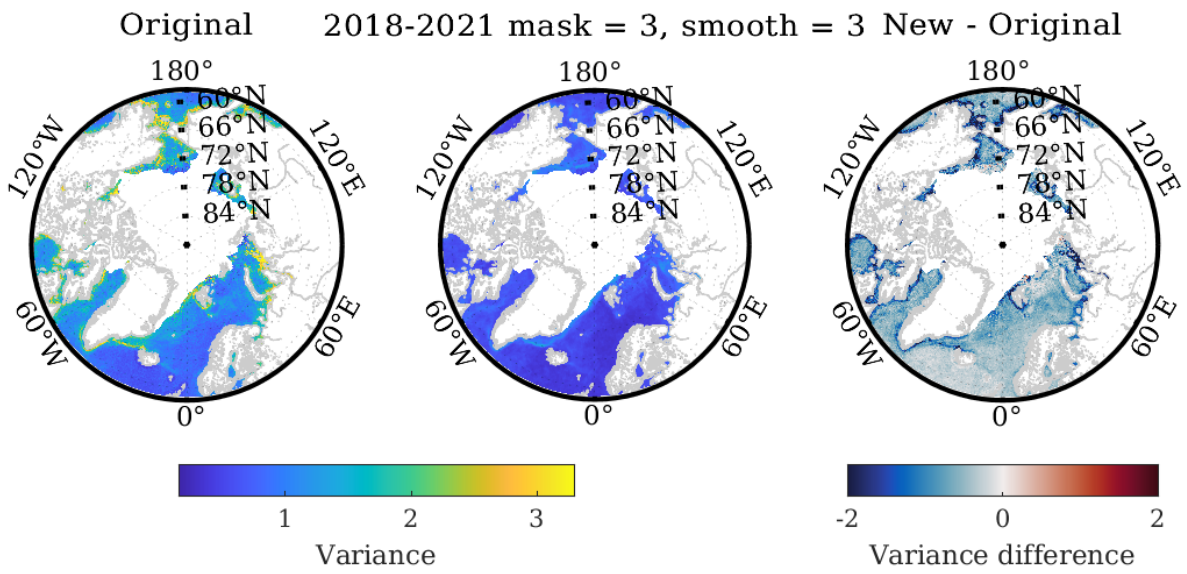

Figure 8. Original (left) and newly estimated (middle) 1st-guess variance fields (in degrees K) and (right) their associated difference (in degrees K).

To assess the impact that the 1st-guess variance field has on the modelled error field provided with the daily L4 analysis, the example date of January 01, 2021 was selected and the error field was re-calculated using the new 1st-guess variance field. Figure 9 shows the original (left) and new error fields (middle) and their associated difference (right). Significant parts of the Baffin Bay, Barents, Bering and Norwegian Sea show a reduction of the estimated errors by approximately 0.5°, indicating that the newly estimated 1st-guess variance field reduces the modelled errors provided with the daily L4 analysis. This indicates that applying the newly proposed 1st-guess variance, will reduce the modelled uncertainties provided with the product.

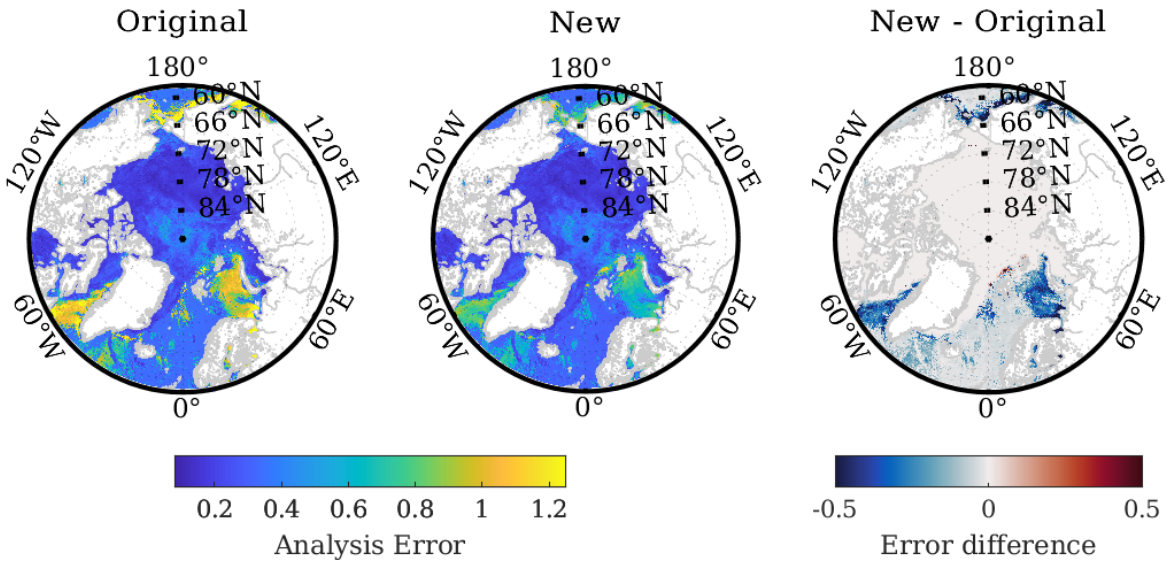

Figure 9. Original (left) and newly estimated (middle) error field and their associated difference (in °K).

To confirm this, the new 1st-guess variance field was used to re-calculate the L4 daily analysis and associated error fields for the test year of 2021 and the modelled uncertainties were validated again using the drifting buoys. Figure 10 shows the uncertainty analysis using the newly calculated L4 SST/IST and error fields with the implementation of the new 1st-guess variance field against drifting buoys. Results can be directly compared to Figure 7.

An overall reduction of modelled errors by approximately 0.46° is observed compared to the original modelled errors, manifested as a decrease in the range of values in the x-axis, for drifting buoys; previous maximum was 1.64° while the current maximum is 1.2°. Furthermore, the standard deviation (vertical lines) of the L4 product vs in situ observations increases with increasing error estimates following better the theoretically expected error (dashed lines), in particular for the modelled errors of around 0.3°, which is where most match-ups are observed (blue line in (b)), but also for the error estimates higher than 0.4°, where fewer matches are present but the error follow the theoretical errors very well.

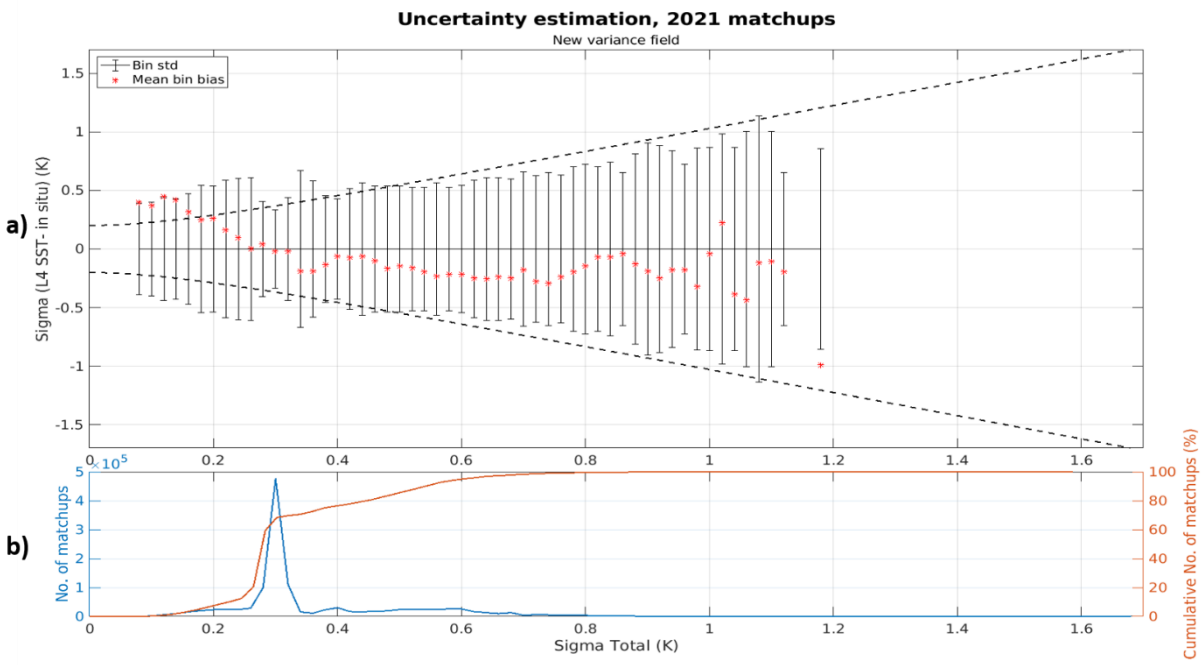

Figure 10. (a): Uncertainties (x-axis) provided with the daily L4 analysis, against the estimated standard deviation of the L4 SST analysis minus in situ SST (y-axis) from drifting buoys. (b) shows the availability of match-ups (blue) and cumulative availability (orange).

To assess the impact of the new vs the old 1st-guess variance field on the daily SST/IST analysis, validation of the SST produced using the old and new 1st-guess variance field, for 2021, was performed against drifting buoys. The results shown in Table 5 indicate almost no change in the statistics for the L4 SST between the two versions, beyond a negligible reduction of the bias in the 3rd decimal and a reduction of the standard deviation in the 2nd decimal. The small decrease in the number of match-ups, may be associated with the fact that almost no match-ups exist for uncertainties higher than 0.7° (blue line) although this was not the case for the original uncertainty analysis shown in Figure 7.

*Table 5. Validation statistics for the L4 SST against drifting buoy measurements for the **Reference** (using old 1st-guess variance) and the **Updated** (using new 1st-guess variance) version for 2021.*

Product	**Variable**	**Mean Bias**	**Standard deviation**	**RMS**	**Nobs**
SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008 **Updated-Drifting buoys**	SST (°C)	0.032	0.48	0.48	1232261
SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008 **Reference-Drifting buoys**	SST (°C)	0.037	0.49	0.49	1232326

IV.5 Assessment of sea ice concentration input data

Version 1.6 of this QUID document presented results showing the impact of updating the Sea Ice Concentration fields (Section IV.3) introduced in 2021 which used the sea ice concentration product OSI-401-b and some additional filtering criteria to remove spurious sea-ice.

In April 2023, the sea ice concentration product OSI-401-b, became obsolete and was automatically substituted by product OSI-401-d[8]. At the same time a sea ice concentration product based on the Advanced Microwave Sensing Radiometer (AMSR) sensors, was also released, i.e. OSI-408-a[9]. Due to many changes of the upstream SIC products, it is necessary to assess the differences of the available products and revise what is being used for the L4 SST/IST production chain, in order to ensure that the best and most accurate SIC upstream product is used to characterise the surface type (open water, MIZ, sea ice). Furthermore, since the AMSR sensors provide higher spatial resolution (15-25 km) compared to the series of the Special Sensor Microwave Imager/Sounder (SSMIS) sensors (30-60 km), this new product is assessed as a replacement to OSI-401-d.

As product OSI-401-b was substituted with OSI-401-d, it was challenging to find an appropriate overlapping period for product comparison and assessment. Therefore, only examples from March 2023 are used, and this is not always the best period to demonstrate issues in the different SIC products arising from different classification of ice-covered vs ice-free areas. Assessment was performed using heatmaps which show the number of pixels in each of the products that for a certain temperature value have a certain SIC value. Ideally, high SIC values should be constraint to low surface temperatures.

Figure 11 shows the heat-maps from the original OSI-401-b unfiltered (ice_conc_unfiltered) product (left) and the version using the additional filters and the 1 km CMEMS SIC for the Baltic region which was used in the operational SST/IST NRT production chain until April 2023 (right). The left figure has a high number of grid points where the SIC is high and the surface temperature is also high which in many cases is unrealistic and caused by land spill-over effects. The right panel with the additional filters applied shows that most of the unrealistic high SIC values over high surface temperatures are removed thus the OSI-401-b+filters version offers a better representation of the real sea ice cover. The horizontal stripes are a result of the sea ice charts used in the Baltic region. In addition, SIC values below 15 % were removed before the extrapolation into fjords was applied.

[8] https://osi-saf.eumetsat.int/products/osi-401-d

[9] https://osi-saf.eumetsat.int/products/osi-408-a

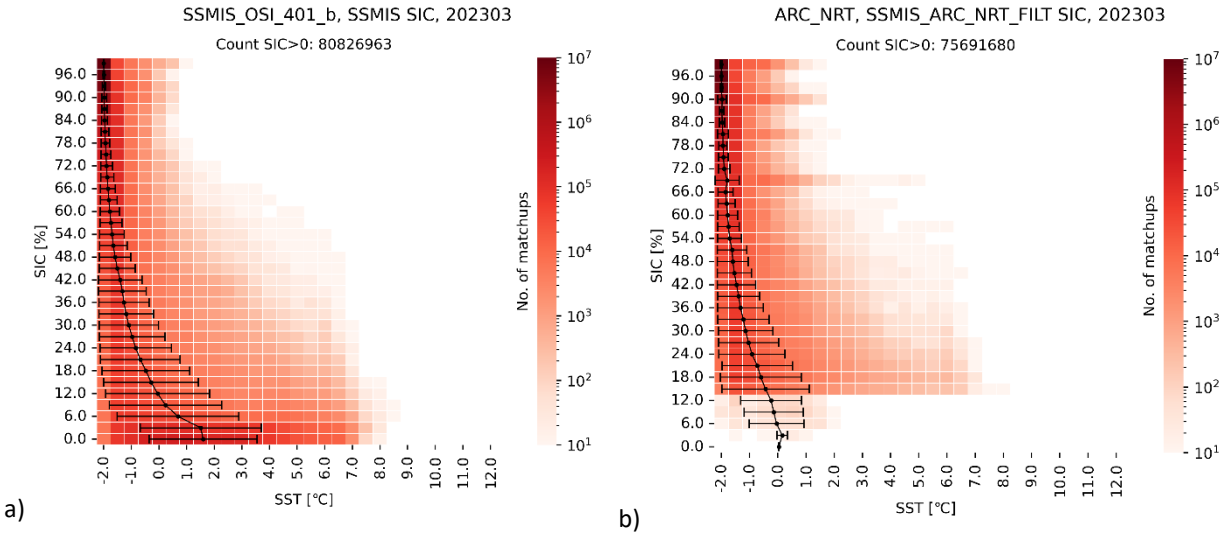

Figure 11. Heat-maps (data collected in March 2023) from (a) the standard OSI-401-b product and (b) the version with additional filters and the 1 km CMEMS SIC for the Baltic region, which was used operationally in the SST/IST NRT production chain until April 2023. The horizontal axis indicates Surface Temperature, the vertical axis indicates Sea Ice Concentration. The units measured in the colour bar are the number of pixels that match-up for each SST/SIC combination.

Top raw (a and b) of Figure 12 show for March 2023, the replacement product OSI-401-d (available from April 2023 onwards as a substitute product to OSI-401-b) with the standard filters applied by the production centre (i.e. variable ice_conc) and no additional filters applied from our side (panel a) and the candidate replacement product OSI-408-a, also with the standard filters applied by the production centre (panel b). Both products show a better representation of the SIC vs SST distribution compared to the original OSI-401-b (Figure 11a) and a similar representation for SIC>15% to the version with the additional filters and the 1 km CMEMS SIC for the Baltic region (Figure 11b). Between the two products, OSI-401-d and OSI-408-a, it is found that the differences are small during March (a and b Figure 12), but a slightly higher number of pixels with SIC between 24% and 50% for which the temperature is between 2 and 6 degrees are identified in OSI-401-d. For July 2023 (bottom row, c and d, in Figure 12), a challenging period where many mis-classifications of real sea-ice can occur, it is found that the OSI-401-d (c) indeed shows more cases with existing sea ice and higher SSTs compared to OSI-408-a (d). This is expected since OSI-401-d is affected more by land spill-over effects due to the coarser resolution of the SSMIS sensors. This is much less the case for OSI-408-a, which in addition also has a higher resolution; this is considered an advantage for the better characterization of SIC closer to the land (coastal areas, enclosed basins, large bays and fjords).

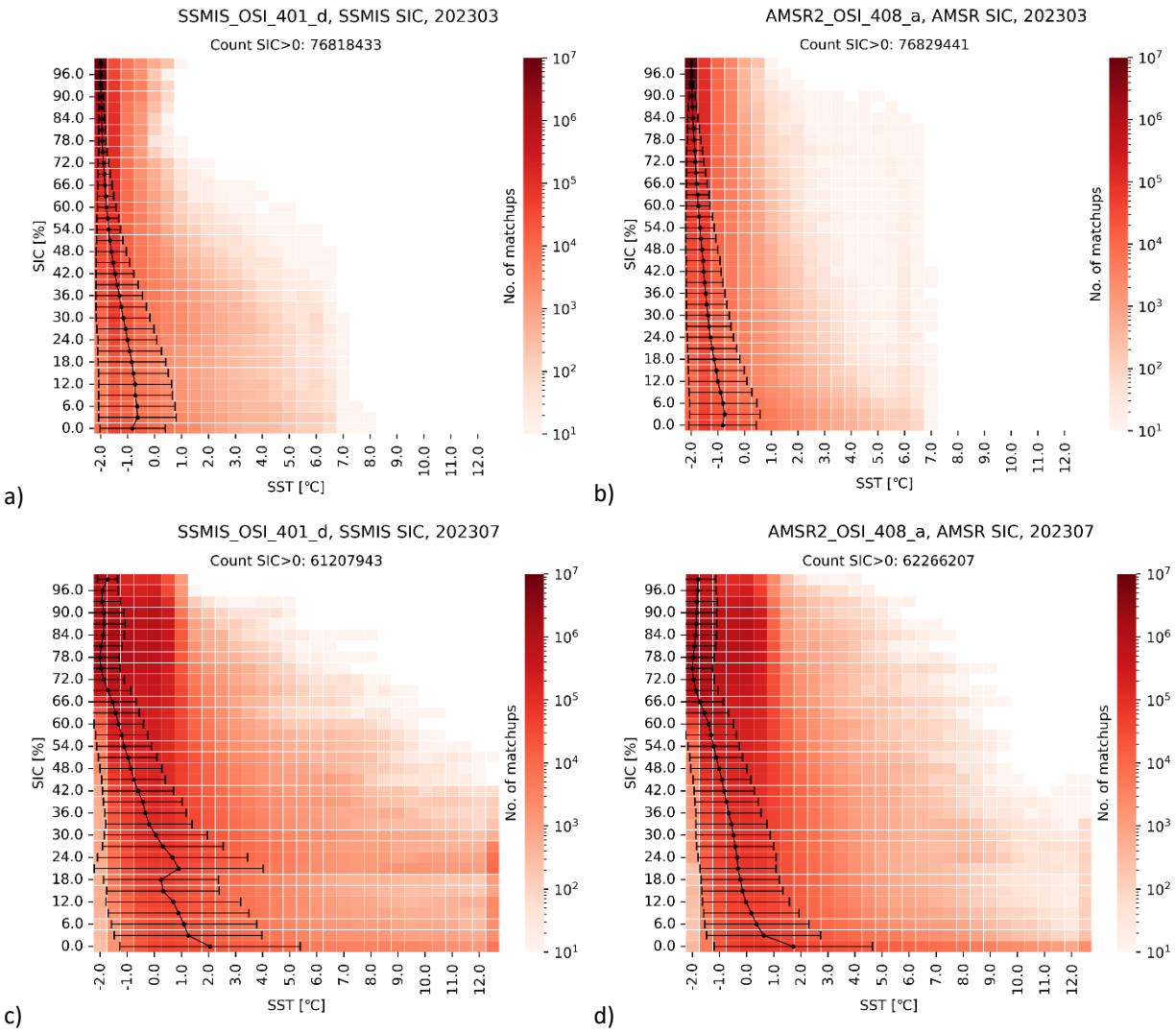

Figure 12. Heat-maps (data collected in March 2023) (top) from (a) the replacement OSI-401-d product released in April 2023 and (b) the OSI-408-a product. Heat-maps (data collected in July 2023) (bottom) from (c) the replacement OSI-401-d product and (d) the OSI-408-a product. The horizontal axis indicates Surface Temperature, the vertical axis indicates Sea Ice Concentration. The units measured in the colour bar are the number of pixels that match-up for each SST/SIC combination.

This higher resolution is illustrated in Figure 13 where the daily SIC field from one day (22/03/2023) of OSI-401-d (a) and OSI-408-a (b) are presented. Because of the improved feature resolution and the less pronounced land spill-over effects, it is suggested to switch to OSI-408-a. Both products still have coastal challenges in the Baltic Sea and for that reason it is suggested to use the high-resolution Baltic 1 km CMEMS SIC product based on manual ice charts in this region. An example of this product is also provided in Figure 13 (c).

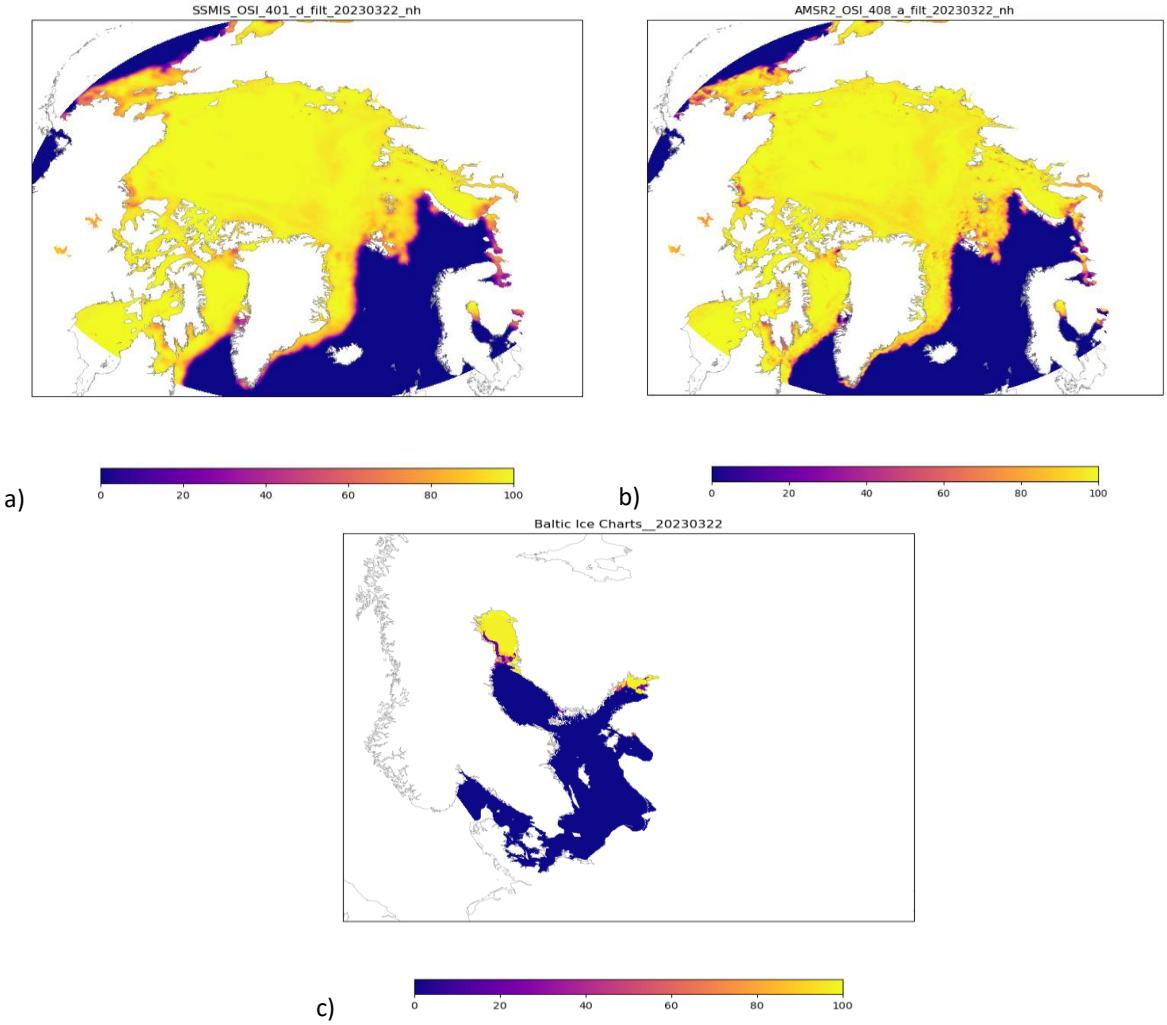

Figure 13. Examples of the SIC field (collected on 22nd March 2023) from the (a) OSI-401-d, (b) the OSI-408-a and (c) the High Resolution Baltic 1 km CMEMS SIC products, respectively. The combination of the OSI-408-a and the HR CMEMS SIC is the proposed evolution for a new SIC field. The colour bar indicates sea ice concentration from zero (dark blue) to 100 percent (light yellow).

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system including changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among others, which may impact product quality.

Date	Change/Event description	System version	Other
15/01/2020	Addition of Sentinel-3B SLSTR SST, NOAA20 VIIRS SST and VIIRS Suomi-NPP SST/IST observations.	Unchanged	Document version 1.5
10/09/2021	New filtering criteria are applied on the input data for the generation of the ice mask field, prior to the OI reconstruction.	Unchanged	Document version 1.6
15/06/2023	New template, routine validation statistics of IST component with SIMB3, analysis/improvement of SST uncertainties	Unchanged	Document version 2.0
31/07/2024	Assessment of sea ice concentration input data	Unchanged	Document version 2.1
09/05/2024	Inclusion of L3S product validation	Unchanged	Document version 3.0

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version. Any quality changes are summarised below.

In v1.5 of the QUID document the Sentinel 3B SLSTR SST, NOAA 20 VIIRS SST and VIIRS_NPP SST/IST observations were introduced. The inclusion of the new data showed improvements in the SST performance, both for bias and standard deviation, when compared to drifting buoys. Similarly, improvements in both bias and standard deviations were observed for the IST performance, when compared against near surface air temperature observations.

In v1.6, new filtering criteria were applied to the input data for the generation of the ice mask field, prior to the OI reconstruction. From analysis during the 3-month period from July to September 2020, it was found that while no significant difference between the two products was identified, there was a marked impact on validation against drifting buoys with the new version demonstrating improved mean difference and standard deviation for the analysed surface temperature parameter.

In v2.0, the new template was used along with the validation of the error_analysis variable included in the daily files. This resulted in changes of the 1st guess variance used to calculate the daily fields and although almost no impact to analysed_st variable is detected, the error_analysis variable was reduced to better represent the real uncertainty. Moreover, new IST in situ data were used for routine validation and these can be seen in Table 2.

VII REFERENCES

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